

Replicating Benchmark Results: How Robust are Benchmark Performances?

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Abstract

NLP benchmarks are an important measure of model comparison and a driving factor for innovation and improving models, but their robustness and replicability remain unclear. This paper investigates the stability and generalizability of benchmark performance by conducting a replication study on the SST-2 task of the GLUE benchmark, using two language models of different sizes, BERT and DistilBERT. The experiment consists of a series of controlled perturbations and variations, including random seeds, different train-validation splits, and subsets of the training data. Results showed that the model performance remained quite stable under these variations, suggesting a strong internal validity. However, evaluating the models on the SST-2 subtask of the Adversarial GLUE dataset, with introduces adversarial attacks and out-of-distribution data, finds a dramatic performance drop of over 60%, suggesting the models lack external validity and robustness to other domains. Thus, benchmark scores should be interpreted with caution and complemented by domain-specific and robustness-oriented evaluation measures.

1 Introduction

Since the rise of transformers (Vaswani et al., 2017), a multitude of language models have been developed with a variety of different parameters, trained on different corpora, and fine-tuned for different downstream tasks. Navigating these models to figure out which is the current state of the art and which is best suited for a particular downstream application is quite a difficult endeavor, so *benchmarks* have become one of the most popular ways of evaluating and comparing language models to each other. Benchmarks are test suites comprised of different tasks that evaluate the abilities of language models with a variety of evaluation metrics

and on a variety of subtasks. Their aim is to provide a platform for comparing language models systematically, so that the state of the art language model is clearly definable as the one that performs best on the benchmarks. This kind of performance-based evaluation is a great incentive for driving innovation in a field, and it promotes the goal of creating better and more accurate models.

However, benchmarks have also been criticized for focusing on single-number performance only, while neglecting other important aspects of building language models: robustness, model size and inference latency, and following from that, computational cost and energy efficiency (Ethayarajh and Jurafsky, 2020). Moreover, benchmark results are often reported as point estimates—accuracy or F1 scores, sometimes from a single run - which can hide substantial variance across random seeds, data splits, or implementation details. This raises questions about the reproducibility and replicability of benchmark results and their external validity, particularly when the models are applied in new domains where there may be adversarial attacks or where the data is not presented as perfectly as in benchmark data sets.

To investigate exactly this replicability of benchmark results and their robustness, this paper will conduct a small-scale replication study focused on the GLUE benchmark (Wang et al., 2018), using DistilBERT (Sanh et al., 2019) and BERT (Devlin et al., 2019) as representative language models. Focusing on the SST-2 subtask of the GLUE benchmark, small perturbations such as random initializations, training on subsets of the data, and different train/validation splits are applied to see how they affect the benchmark results. Furthermore, to provide even more severe testing, the SST-2 subtask from the Adversarial GLUE (Wang et al., 2022) benchmark is used to test how well the models can deal with adversarial attacks and out-of-distribution data.

Section 2 of this paper discusses the replication crisis and problems with many popular benchmarks, while section 3 describes in detail the experiments to test the GLUE benchmark, describing the models used and the different perturbations applied. Section 4 presents the results of the perturbations and tests, with section 5 discussing the findings and drawing important conclusions from the data. Section 6 presents limitations of the present study and provides an outlook to the future of benchmarks, with section 7 concluding the paper with the main takeaways.

2 The Benchmark & Replication Crisis

2.1 The GLUE Benchmark

One of the most common benchmarks to test the language understanding skills of language models is GLUE, the General Language Understanding Evaluation benchmark (Wang et al., 2018). GLUE consists of nine different sentence or sentence-pair tasks for natural language understanding, ranging from sentiment analysis to recognizing entailments or selecting correct pronoun referents (Wang et al., 2018). The authors claim that to achieve good performance on the benchmark, a model needs to share substantial amounts of knowledge across all tasks (Wang et al., 2018). GLUE has been extended to SuperGLUE, a new benchmark extending GLUE with more difficult and diverse tasks to allow for more challenging testing of language models (Wang et al., 2019).

2.2 The Problem with Benchmarks

Benchmarks like GLUE have been a driving force of innovation and progress in NLP, but they drive progress solely focused on "performance-based evaluation" (Ethayarajh and Jurafsky, 2020). This means that most common NLP benchmarks like GLUE evaluate and rank models based on accuracy, precision, recall or F1 score, neglecting other important qualities of language models, such as compactness, fairness, or efficiency (Ethayarajh and Jurafsky, 2020). This results in NLP engineers solely focusing on models with a higher performance, to achieve higher ranks in benchmark leaderboards.

Nonetheless, when deploying models for real-world applications, accuracy is not everything: for real-world use, models should be as compact and energy-efficient as possible, and they should be robust to adversarial examples and to out-of-distribution data (Ethayarajh and Jurafsky, 2020).

There is thus a significant mismatch between what is prioritized by benchmark leaderboards and what is important in the industry: the top-ranked model in a benchmark may be entirely unusable in the industry because it is too costly or does not generalize well to unseen data (Ethayarajh and Jurafsky, 2020).

Leaderboards should thus encourage transparency and include other metrics such as cost and robustness in their ranking function. Ethayarajh and Jurafsky (2020) suggest a dynamic, customizable leaderboard that allows weighing of the metrics to allow prioritization of accuracy, cost or robustness (or other metrics). This would allow models that sacrifice accuracy for a better robustness or a lower cost to achieve higher rankings in a leaderboard that prioritizes these metrics.

Some benchmarks have made a start at encouraging practices like building robust models, for instance Adversarial GLUE, which introduces adversarial attacks and different kinds of perturbations to the GLUE dataset (Wang et al., 2022). They include a multitude of perturbations from word-level to sentence-level changes as well as human-crafted adversarial examples to test if models can understand numerical reasoning or the order of events (Wang et al., 2022). They found that most common language models like BERT, RoBERTa etc. performed much worse on the adversarial GLUE benchmark than on the original GLUE, showing that the perturbations significantly affected performance (Wang et al., 2022). Other newer benchmarks such as the one presented by Bogavelli et al. (2026) also include a variety of perturbations to test LLM robustness, finding that a lot of models are least robust to cross- or multilingual perturbations, as well as positional perturbations.

3 Testing the GLUE Benchmark

The following chapter will perform rigorous tests to one task of the GLUE benchmark to discuss how generalizable the result of the benchmark is, and whether the models that perform well on the benchmark are robust to perturbations and adversarial attacks. For simplicity and due to hardware constraints, this paper will focus on only one of the GLUE tasks, namely SST-2, which is commonly used to evaluate language models. SST-2, the Stanford Sentiment Treebank, tests if models can perform sentiment analysis on sentences from movie reviews which were annotated by humans on

whether they contain positive or negative sentiment (Wang et al., 2018).

3.1 Models

3.1.1 BERT

The first model used to replicate and test the GLUE benchmark is BERT (Devlin et al., 2019). BERT is one of the most commonly used transformer-based encoder models that produces deep contextual embeddings of texts, and it is used widely for fine-tuning for a variety of downstream tasks. Devlin et al. (2019) employ the GLUE benchmark to evaluate BERT’s NLU capabilities, reporting an average of 79.6 for BERT-base, with an accuracy of 93.5 for SST-2 (Devlin et al., 2019). BERT has since become a widely used baseline for text classification and other NLU tasks, serving as a reference point for subsequent model architectures, sparking an entire family of BERT models.

3.1.2 DistilBERT

The other model that will be used for replicating the GLUE benchmark results will be DistilBERT (Sanh et al., 2019). DistilBERT is a smaller, distilled version of BERT that was trained using knowledge distillation from the teacher model, BERT (Devlin et al., 2019). DistilBERT reduces the number of layers of BERT by a factor of two, leading to an overall 40% smaller transformer model (Sanh et al., 2019). Sanh et al. (2019) state that their model retains about 97% of the performance of BERT, despite being more compact and being much faster at inference. They assess the performance on the GLUE benchmark, reporting the development set scores of each GLUE task and comparing the scores to BERT (Devlin et al., 2019) and ELMo (Peters et al., 2018). In their evaluation, DistilBERT only lingers slightly behind BERT, and outperforms ELMo for most GLUE tasks.

3.2 Baseline

In order to establish a baseline, the first run aims at closely replicating the SST-2 scores reported in the dataset papers for both BERT and DistilBERT, based on one run per model. The models were fine-tuned with seed = 42 and a learning rate of $2e-5$ for 3 epochs. For BERT, a batch size of 16 for training and 32 for evaluation was used, whereas for DistilBERT a batch size of 32 for training and 64 for evaluation was selected due to the smaller model architecture.

3.3 Perturbation 1: Random initialization with different seeds

To ensure the stability of the accuracy, each model was then evaluated on the SST-2 task using five different seeds (1, 11, 21, 31, 42) to compare how accuracy might fluctuate across evaluation runs. The training arguments and parameters used were the same as in the baseline, with the only variable changing being the seed. This variation changes the random initialization of weights and the shuffling of data batches during training, which can introduce small stochastic differences even under identical hyperparameter settings. By keeping all other factors constant, this perturbation isolates the effect of randomness in the optimization process on the final accuracy, allowing an estimation of the variance and reliability of reported benchmark scores.

3.4 Perturbation 2: Training on subsets of the data

To investigate how data quantity affects the model robustness, each model was then fine-tuned on randomly sampled subsets of the SST-2 training set, once using 50% and once using 25% of the dataset. All other training parameters were held constant to the baseline. This perturbation shows how reliant the model’s accuracy is on the size of the benchmark dataset, and whether smaller amounts of training data lead to a large decrease in accuracy.

3.5 Perturbation 3: Using different train / eval splits

To confirm that the accuracy values are not dependent on the particular split of training and evaluation data of the benchmark, each model was further fine-tuned on new partitions of the SST-2 dataset. The dataset was divided into train and validation set using three different random split seeds (7, 17, 27). The size of the splits were kept the same, 90% for training, 10% for validation, only the distribution of the samples to the splits was changed. This perturbation investigates whether the results are dependent on the particular data split of the benchmark, or if the model will generalize to new data splits just as well.

3.6 Adversarial GLUE vs. GLUE

In a final run, both models were tested on the SST-2 task from the Adversarial GLUE benchmark (Wang et al., 2022). As described in section 2.2, Adversarial GLUE introduces different kinds of adversarial

Setting	DistilBERT	BERT-base
Baseline (seed 42)	0.906	0.925
Random seeds (mean $\pm$ sd)	0.907 ± 0.004	0.927 ± 0.003
Subset 50%	0.903	0.924
Subset 25%	0.904	0.920
Alt. split (seed 27)	0.948	0.951
AdvGLUE (dev)	0.291	0.304

Table 1: Accuracy on GLUE SST-2 (dev) and Adversarial GLUE (dev) for DistilBERT and BERT-base under controlled perturbations. All values correspond to accuracy, the random seed results show mean $\pm$ standard deviation across five runs.

attacks into the dataset to make it more difficult for language models to solve the tasks. Since Adversarial GLUE is based on the same dataset as GLUE, a direct comparison between the two scores is possible and allows for a stress test for model generalization, to see if the model can perform the same task on a more difficult dataset with adversarially modified texts.

4 Results

Table 1 presents the results of all experiments. The baseline run with seed 42 achieved an accuracy of 0.925 for BERT-base and 0.906 for DistilBERT. [Devlin et al. \(2019\)](#) reported an accuracy of 93.5% for BERT-base (although they evaluated on the official hidden test set, whereas the experiments here were done on the validation set, so the results are not entirely comparable). [Sanh et al. \(2019\)](#) reported an accuracy of 91.3% for DistilBERT on the SST-2 validation set in their paper, as an average of five runs with different seeds. This is comparable to this paper’s perturbation 1, where the median came to 0.906 for DistilBERT, with a mean of 0.907.

Fine-tuning on a subset of the data did not seem to have a large impact for neither DistilBERT nor BERT: DistilBERT achieved an accuracy of 0.903 on half of the training data, and 0.904 on a quarter of the data, whereas BERT achieved 0.924 and 0.920 respectively.

Choosing an alternative data split improved results for both models, with seed 27 achieving 0.948 for DistilBERT and 0.951 for BERT.

The largest discrepancy occurred on the evaluation on adversarial GLUE: both models showed a significant decrease in accuracy on the adversarial benchmark, with DistilBERT achieving only 0.291 (67.9% relative performance drop) and BERT achieving 0.304 (62.14% relative performance drop).

5 Discussion

The presented experiments performed a series of stress tests to check if the SST-2 task of the GLUE benchmark delivers generalizable results. Small random perturbations like changing the seed only minorly affected the accuracy, which shows that the benchmark result is not contingent on a singular seed or a singular "best run". Across all five seeds that were tested, the accuracy was always above 0.90 for DistilBERT and above 0.93 for BERT, showing a solid internal validity. Every seed also produced a similar accuracy to the accuracies reported in the dataset paper, showing an easy replicability of the results.

Furthermore, training on only a subset of the dataset did not harm performance by much (if at all). Even on only 25% of the training dataset, the accuracy still remained above 0.90 for DistilBERT and above 0.92 for BERT, showing that the models are able to correctly perform sentiment analysis even with very little training. This confirms that the benchmark results are not contingent on the dataset size: even with smaller amounts of training data, similar results can be achieved.

Changing the split between training and validation did not harm the performance either - on the contrary, an alternate split with seed 27 even improved both DistilBERT’s as well as BERT’s accuracies to 0.948 and 0.951 respectively. This also confirmed that the benchmark results are not due to a particularly "lucky" data split, but that other splits perform similarly or even better than the benchmark split.

The only perturbation that significantly harmed model performance was using the Adversarial GLUE benchmark. Both models experienced a dramatic drop in accuracy: from 0.906 to 0.291 for DistilBERT and from 0.925 to 0.304 for BERT. This shows a relative drop of over 60% for both models, which is quite drastic: both models perform below chance on the adversarial dataset, showing that they are not just failing to generalize, but are probably systematically misclassifying examples. This is very likely a result of the adversarial attacks in the dataset: these are made to be more difficult for the model and are meant to "trick" it into misclassifying, which seems to be successful, as the models seem to be unable to discover these adversarial attacks. This could indicate that the models rely on very superficial syntactic and semantic cues, which the adversarial GLUE likely

exploits, causing the models to make systematically incorrect predictions.

Does this prove that the models are not generalizable? Only somewhat. The smaller perturbations showed that performance on the benchmark does not seem to be contingent on the specific GLUE benchmark size and split, but the adversarial GLUE benchmark seems to show that the models are not very robust to examples that are less straightforward than the GLUE examples. It is very likely that the models are reliant on basic syntactic or semantic cues in the GLUE benchmark (such as negations, or specific positive or negative lexical items) that can be exploited in adversarial examples to mislead the model. Thus, success on GLUE does not guarantee robustness to perturbed or adversarially generated inputs - the benchmark does not guarantee external validity. However, whether the model is generalizable or not depends on the specific task and domain it is meant to be used in: in most simple domains, it will probably be sufficient if the model performs well on GLUE. However, in a domain sensitive to adversarial attacks or perturbations, this may not be sufficient, leading the model to misclassify. Therefore, models necessarily need to be tailored to the domain they are meant to be used in. So, benchmark results should not be taken as ground truth accuracy that applies to all domains, but as a domain-specific accuracy that is not generalizable to all applications.

6 Limitations & Outlook

The experiments in this paper are subject to some limitations and only provide a very small window into the current situation of benchmarks in NLP: for one, both models tested in this paper, BERT and DistilBERT, are models from 2019, and LLMs have developed massively and come a very long way since then. With new models that have improved greatly and exhibit ever more language capabilities, many new problems of benchmarking arise.

First of all, new models have started to be so good that they've reached the ceiling of some benchmarks, and model creation is outpacing the creation of new adequate benchmarks (Gevers et al., 2025). Furthermore, some of the popular benchmarks may have fallen subject to data contamination, with their datasets possibly being used for training of new LLMs. This, and other cases of "shortcut learning", where the LLM uses unrelated patterns in the data to get to the correct answer,

can lead to inflated scores which do not actually tell us anything about how good the LLM is at performing these tasks in downstream applications (Gevers et al., 2025). This leads to quite a large disconnect between the benchmark performance and the actual applicability of the models in real-world tasks. To mitigate these issues, Gevers et al. (2025) present a checklist for both the creation of new benchmarks as well as the usage of benchmarks for model evaluation, ranging from dynamic updates to benchmarks to prevent memorization, to encrypting test sets and keeping them a secret to prevent contamination.

Further, as stated before, models should always be tested in the domain and on the specific downstream task that they are meant to be used in, to ensure that the model can deal with real cases on the data. For this, Gevers et al. (2025) propose some small, domain-specific test suites, to ensure that the good performance on the benchmark is complemented and correlated with a solid downstream performance as well. These test suites should be small, more unique and challenging cases that can help in qualitatively analyzing the LLMs behavior.

7 Conclusion

This paper conducted experiments to test the reliability of the SST-2 subtask of the popular GLUE benchmark. These experiments showed that the SST-2 task seems to have a very high internal validity, with small perturbations to dataset size, seed, or train and validation splits having a very minimal effect on accuracy. However, using the SST-2 task of the adversarial GLUE benchmark showed that introducing larger perturbations and adversarial attacks into the validation set leads to a drastic drop in accuracies. This shows that models that perform well on the GLUE benchmark are not necessarily robust models and may still fall victim to adversarial attacks and out-of-distribution examples.

This confirms that while benchmarks drive innovation and can lead to improvements, they do not assess all of the necessary qualities that models should possess for real-world applications. Models should, depending on the domain where they will be employed, also be robust, generalizable to out-of-distribution data, and compact. To drive innovation in these aspects, benchmarks like Adversarial GLUE are very needed and relevant to prioritize robustness and generalizability in benchmarks, shared tasks, and NLP overall. Furthermore,

additional testing on small, domain-specific test suites can help evaluate models to choose the best performing model for a certain downstream application. While benchmarks are an important measure of comparison between models, their leaderboards and scores should always be evaluated critically.

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